

# An Investigation of the Period of a Pendulum

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This paper presents an experimental investigation into the period of a simple pendulum, validating its dependence on initial angle ( $\theta_0$ ) and length ( $L$ ), while confirming independence from mass ( $m$ ). Using dimensional analysis and physical arguments, the period is predicted to be  $T = \sqrt{L/g} \cdot f(\theta_0)$ , with the function  $f(\theta_0)$  approximated by the series expansion  $f(\theta_0) \approx A + B\theta_0^2$ . Data were collected for five pendulums of varying length and mass using video analysis, and parameters were extracted using Orthogonal Distance Regression (ODR) to account for uncertainties in both axes. The results strongly support the theoretical model. The weighted mean of the zeroth-order term was found to be  $\bar{A} = 6.2850 \pm 0.0025$ , agreeing with the theoretical value of  $2\pi$  within  $0.72\sigma$ . The first-order coefficient was  $\bar{B} = 0.3973 \pm 0.0102 \text{ rad}^{-2}$ , consistent with the theoretical  $2\pi/16$  within  $0.45\sigma$ . Mass independence was confirmed by comparing 37 g and 6 g bobs, which yielded statistically indistinguishable parameters. All five normalized period functions collapsed onto a single universal curve, demonstrating that  $f(\theta)$  is independent of both mass and length. Furthermore, the small-angle period dependence on length was fit to  $T_0 = C \cdot L^P$ , yielding an exponent  $P = 0.501 \pm 0.001$  ( $1.0\sigma$  from theory) and constant  $C = 2.009 \pm 0.002 \text{ s/m}^{0.5}$  ( $1.3\sigma$  from theory), confirming the  $T_0 \propto \sqrt{L}$  relationship.

## I. INTRODUCTION

The harmonic oscillator is a foundational model in physics, describing systems that experience a restoring force linearly proportional to displacement from equilibrium. From atomic vibrations in crystal lattices to resonant circuits in electronics, many physical phenomena reduce to harmonic motion. The simple pendulum—a point mass suspended on a massless rod—is a classical example, often serving as a first introduction to oscillatory behavior.

Despite the idealized pendulum's well-established theory, experimental reproducibility is essential to confirm results in science. This work systematically validates the pendulum model by measuring period dependencies on initial angle ( $\theta_0$ ), length ( $L$ ), and mass ( $m$ ) across five different configurations.

### A. Theory and Model Development

Pictured in Figure 1 is my experimental apparatus. Three stands with vertical bars were placed on a table – two supporting a horizontal bar to which the pivot of the pendulum attached, and which was perpendicular to the third vertical bar which supported the protractor behind the pendulum string. A detailed discussion on why this setup was chosen can be found in the "Methods" section.

Before any data was taken, time was taken to determine what parameters could affect the period of a pendulum and how, through methods of dimensional analysis combined with physical arguments. The parameters are are:

- The gravitational constant ( $g$ ) [1].

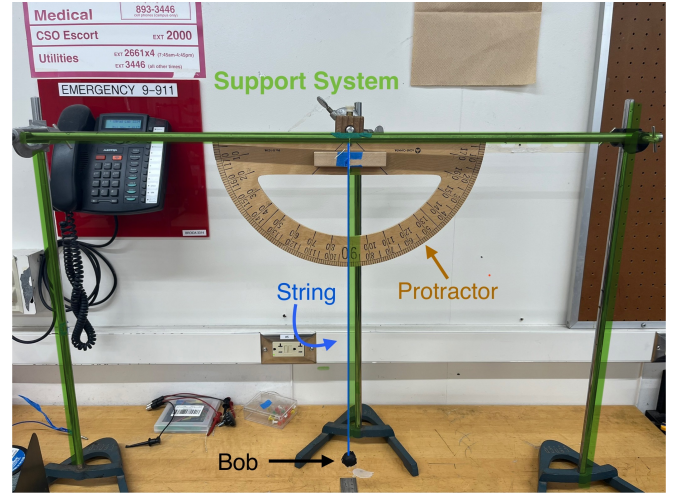


FIG. 1. Experimental setup showing pendulum configuration. An I-phone which recorded the motion using a slow motion camera was positioned in front of the setup (not pictured)

- Parameters characterizing air resistance: density ( $\rho$ ), bob area ( $S$ ), and velocity ( $v$ ).
- The mass of the bob ( $m$ ).
- The initial angle of release ( $\theta_0$ ).

These parameters determine the resulting Period ( $T$ ). It was assumed that air resistance would be proportional to the surface area ( $S$ ) of the bob and to the velocity ( $v$ ) such that an increase in either of these variables would increase the effect of drag, i.e.,

$$F_{\text{drag}} \propto S^\alpha v^\beta; \alpha, \beta > 0 \quad (1)$$

We expect the the mass doesn't affect the period of oscillations, in the conditions of our experiments (negli-

gible drag). None of the relevant quantities— $g$ ,  $l$ , or  $\theta_0$ —contain mass in their dimensions, and there is no way to combine Length and Time units to produce a Mass unit. The only mechanism that could introduce mass dependence would be a dissipative force such as air resistance, where the drag force would need to be compared against the gravitational restoring force. Such a comparison would necessarily involve the bob’s mass. However, by using relatively heavy bobs and operating at moderate velocities, we ensured that air drag remained negligible throughout our measurements, justifying its exclusion from our model. Later (Goal 3), we will test this mass-independence prediction directly by comparing pendulums of identical length but vastly different masses.

We have four parameters:  $T$ ,  $l$ ,  $g$ , and  $\theta_0$  (which is already dimensionless). My goal was to write the most general dimensionless equation relating these parameters. Using Buckingham’s Pi theorem [2], which states that the number of independent dimensionless groups equals the number of parameters ( $k$ ) minus the number of fundamental dimensions ( $r$ ). In our case, we have  $k = 4$  parameters ( $T, l, g, \theta_0$ ),  $r = 2$  fundamental dimensions (Length, Time), and 3 dimensionful parameters ( $T, l, g$ ). This implies  $k - r = 2$  independent dimensionless groups. We can choose  $\pi_1 = \theta_0$  and  $\pi_2 = T\sqrt{g/l}$ . The theorem states that any physical relationship must be of the form  $F(\pi_1, \pi_2) = 0$ , which can be rewritten as  $\pi_2 = f(\pi_1)$ . This yields the most general relation:

$$T = \sqrt{\frac{l}{g}} f(\theta_0) \quad (2)$$

There were three goals to the following experiments:

1. Find this function  $f(\theta_0)$  and show that it is consistent throughout all pendulums, regardless of length or mass.
2. Confirm the  $T \propto \sqrt{l}$  relationship.
3. Show that varying the mass  $m$  does not affect the period.

Next, we assume that the functional form for  $f(\theta_0)$  can be a Taylor expansion about that equilibrium point:

$$f(\theta_0) = \sum_{n=0}^{\infty} c_n \theta_0^n = c_0 + c_1 \theta_0 + c_2 \theta_0^2 + c_3 \theta_0^3 + \dots \quad (3)$$

Looking at Eq. (3), we can further constrain what  $f(\theta_0)$  should be. We know from the physical symmetry of the system that the period of the pendulum is the same whether we start from  $+\theta_0$  or  $-\theta_0$ . That is, whether we release the bob from the same initial angle to the left or right side of the protractor should not change the period. Thus,  $f(\theta_0)$  must be an even function, i.e.,  $f(\theta_0) = f(-\theta_0)$ . This physical constraint requires that all coefficients of odd powers ( $c_1, c_3, \dots$ ) in Eq. (3) must be zero.

This simplifies our expansion to:

$$f(\theta_0) = c_0 + c_2 \theta_0^2 + c_4 \theta_0^4 + \dots \quad (4)$$

We can truncate this series after the second-order term, though this approximation has limitations. At our maximum angle of  $\sim 45^\circ$  ( $\approx 0.78$  rad), the ratio  $\theta_0^4/\theta_0^2 \approx 0.61$  means the fourth-order term could contribute significantly if  $c_4 \sim c_2$ . However, our goal is to test whether a quadratic model captures the dominant angular dependence—thus we retain only the second-order term. We are left with the final form for our function:

$$f(\theta_0) \approx A + B\theta_0^2 \quad (5)$$

where  $A$  and  $B$  are the constants  $c_0$  and  $c_2$  to be determined by our fit.

Now we have our final functional form for the period:

$$T = \sqrt{\frac{l}{g}} (A + B\theta_0^2) \quad (6)$$

The exact theoretical solution for the period of a simple pendulum is given by a complete elliptic integral [3], which can be expressed as the series [2]:

$$T = 2\pi \sqrt{\frac{l}{g}} \left[ 1 + \frac{1}{16} \theta_0^2 + \frac{11}{3072} \theta_0^4 + \dots \right] \quad (7)$$

Truncating this series to the same order as our model Eq. (7) gives the theoretical prediction for our normalized period,  $f_{\text{theory}}(\theta)$ :

$$f_{\text{theory}}(\theta) = 2\pi \left( 1 + \frac{1}{16} \theta_0^2 \right) \approx 6.2832 + 0.3927 \theta_0^2 \quad (8)$$

This predicts that our fitted parameter  $A$  should be consistent with  $2\pi$  and  $B$  should be consistent with  $2\pi/16$ .

## II. METHODS

The primary goal of the following experiments is to measure the parameters  $A$  and  $B$  that define this function,  $f(\theta_0)$ . To do this, five pendulums of varying length and mass were tested, each from an initial  $\theta_0$  of approximately  $10^\circ$  to  $45^\circ$  (see Table I).

TABLE I. Pendulums used

| Trial | Length (cm) | Mass (g) |
|-------|-------------|----------|
| 1     | 16.8        | 36       |
| 2     | 26.4        | 36       |
| 3     | 34.7        | 36       |
| 4     | 51.0        | 37       |
| 5     | 50.8        | 6        |

The two different masses used for experiment are pictured below. The two masses were assumed to be of

roughly the same shape. The wooden sphere was slightly larger, which would mean it would be affected by drag more so than the smaller heavier bob. This was shown to be true, as the initial drop angle for a pendulum of similar length decayed more rapidly for the larger wooden sphere than its smaller metallic counterpart.

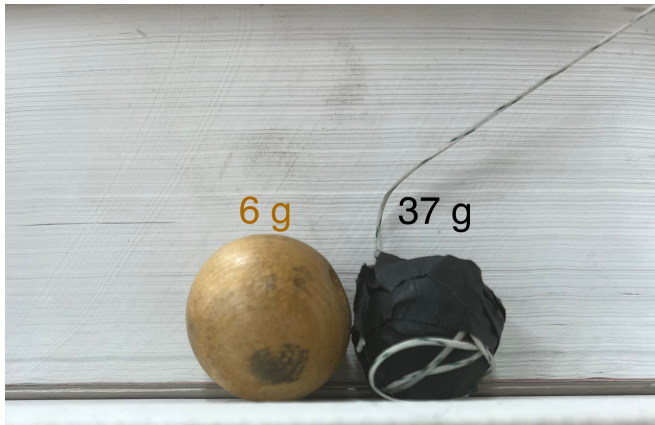


FIG. 2. 6 gram wooden mass compared with 36 g metal mass (with enough tape, the mass became 37 g)

### A. Apparatus and Error Mitigation

Initial tests of various apparatuses revealed a significant source of potential error: deviation from 2-D motion. The final setup (pictured in Fig. 1) was determined through an iterative process to mitigate this deviation. What would happen was that the bob would start its motion in a plane that was parallel to the plane of the protractor, and then over the span of a couple oscillations, it would start moving into or out of this plane, almost as if it were precessing. In the final configuration, a horizontal bar supports the pivot point, which is formed by looping the string around the bar and securing it with tape. The pendulum bob then swings in a plane parallel to this support bar.

Preliminary setups, such as those where the pivot was oriented 90 degrees to this configuration (i.e., where the normal vector of the "ring" of the pivot was perpendicular to the plane of motion), were highly susceptible to uncontrolled precession. The chosen setup minimizes the excitation of this mode of oscillation, effectively constraining the motion to the desired 2D plane and allowing for stable oscillations.

### B. Observational Alignment

Even with this improved physical setup, deviation from 2-D motion could still be introduced by an imperfect release, and apparent errors could be introduced by observational misalignment. Before data collection, it was

essential to align the protractor parallel to the desired plane of motion. Furthermore, the camera was positioned to capture this plane head-on, with no horizontal or vertical angle, to avoid error when analyzing the slow-motion videos.

Proper alignment was an iterative process: the pendulum was released, a test video was recorded, and the video was then reviewed to ensure the plane of motion was parallel to the protractor and the camera's focal plane. Misalignment was evident if a discrepancy appeared between the release angle ( $\theta_0$ ) and the angle at the opposing apex, indicating that the bob was moving out of the 2D plane (i.e., "into the page").

## C. Period vs. Angle Analysis

### 1. Measurement Procedure

For a pendulum of fixed mass and length  $L$ , the period  $T$  was measured for a series of initial angles. These measurements were recorded using a slow-motion video camera. For a given experimental run, the pendulum was released from an initial angle  $\theta_0$  at time  $t_0$ . After  $N$  full oscillations, the pendulum lost some of its energy due to friction, and a new angle  $\theta_1$  ( $\theta_1 < \theta_0$ ) and the corresponding time  $t_1$  ( $t_1 > t_0$ ) were recorded by advancing the video frame by frame.

An experimental run would be voided and restarted if, upon release, the bob was observed to deviate from the desired plane of motion. Most runs required several attempts; however, once a clean in-plane release was achieved, the bob could oscillate for dozens of cycles without significant observable deviation.

To account for error caused by a decrease in initial angle due to air resistance, the representative angle for the run was taken as the mean of the initial and final angles:

$$\theta = \frac{\theta_0 + \theta_1}{2} \quad (9)$$

The period  $T$  for the run was then calculated from the total elapsed time  $\Delta t = t_1 - t_0$  and the number of oscillations  $N$ :

$$T = \frac{t_1 - t_0}{N} \quad (10)$$

### 2. Error Calculation for Angle and Period

Uncertainties were carefully propagated for both angle and period measurements.

**Angle Error ( $\delta\theta$ ):** The protractor used to measure angles had a resolution of  $1.0^\circ$ . We assigned a reading uncertainty of  $\delta\theta_{\text{read}} = 0.5^\circ$  to each of the two independent angle measurements,  $\theta_0$  and  $\theta_1$ . The error in the mean

angle  $\theta$  was then calculated by adding these independent errors in quadrature:

$$\delta\theta = \sqrt{\left(\frac{\partial\theta}{\partial\theta_0}\delta\theta_0\right)^2 + \left(\frac{\partial\theta}{\partial\theta_1}\delta\theta_1\right)^2} \quad (11)$$

[4] Given  $\delta\theta_0 = \delta\theta_1 = 0.5^\circ$ , this simplifies to:

$$\delta\theta = \sqrt{\left(\frac{1}{2} \cdot 0.5^\circ\right)^2 + \left(\frac{1}{2} \cdot 0.5^\circ\right)^2} = \frac{0.5^\circ}{\sqrt{2}} \approx 0.354^\circ \quad (12)$$

**Period Error ( $\delta T$ ):** The video frame rate allowed time resolution to 0.01 s. The “turning point” where timing was stopped usually occurred in a 0.04 s range. Thus, a more conservative instrumental uncertainty of  $\delta t_{\text{inst}} = 0.02$  s was assigned to each time reading ( $t_0$  and  $t_1$ ) to account for this error in identifying the exact turning point of the bob. In reviewing plots of Period vs. Initial angle, it is perhaps evident that this error estimate was *too* conservative as the resulting error bars are noticeably larger than the scatter of the data (see Results below). The error in the total elapsed time  $\Delta t$  was found by adding these two reading errors in quadrature:

$$\delta(\Delta t) = \sqrt{(\delta t_0)^2 + (\delta t_1)^2} = \sqrt{2(0.02 \text{ s})^2} \approx 0.028 \text{ s} \quad (13)$$

[4] The final, total error  $\delta T$  for each data point is therefore

$$\delta T = \delta(\Delta t)/N. \quad (14)$$

Where more oscillations means smaller uncertainty in the period.

There is an evident trade-off in our measurement strategy. Using fewer oscillations reduces the elapsed time  $t_1 - t_0$ , increasing the relative uncertainty in the period. Conversely, measuring over many oscillations allows the amplitude to decay significantly, increasing the uncertainty in the initial angle. An experimental choice was made to balance these competing effects: we allowed the pendulum to oscillate until the amplitude decreased by an amount comparable to our protractor reading precision ( $\sim 0.5$ ). For larger initial angles, this typically corresponded to 3–4 oscillations, while for smaller angles, we measured over 5–7 oscillations before significant amplitude decay occurred.

### 3. Parameter Fitting with ODR

Our physical model,  $T = \sqrt{L/g} \cdot (A + B\theta^2)$ , was linearized to the form  $y = A' + B'x$  by setting  $y = T$  and  $x = \theta^2$ . From our error analysis, we have a well-defined uncertainty for our y-variable,  $\delta y = \delta T$ . Crucially, we also have a non-zero, propagated uncertainty for our x-variable:

$$\delta x = \delta(\theta^2) = \left| \frac{d(\theta^2)}{d\theta} \right| \delta\theta = |2\theta| \delta\theta \quad (15)$$

[4] Standard weighted least-squares routines, such as `scipy.optimize.curve_fit`[7], are insufficient for this analysis. These routines minimize the weighted sum of squared residuals, assuming that the “true” value of the independent variable is the measured value ( $x_{\text{true},i} = x_i$ ) and that all uncertainty lies solely in the dependent variable. As a result, such a fit only closes the vertical gaps between data points and the model, incorrectly ignoring the significant propagated uncertainty in the independent variable.

To properly account for uncertainties in both variables, we employed Orthogonal Distance Regression (ODR), as implemented in the `scipy.odr` package[7]. ODR is a more general and computationally intensive method that does not assume the measured  $x_i$  values are perfectly known. Instead, ODR determines the set of model parameters (e.g.,  $A$  and  $B$ ) that defines the best-fit curve *while simultaneously* determining, for each measured data point ( $x_i, y_i$ ), an “idealized” point

$$(x_{\text{true},i}, y_{\text{true},i}) = (x_{\text{true},i}, f(x_{\text{true},i}; A, B)).$$

These idealized points are additional variables introduced in the fit, with the strict constraint that their  $y$ -coordinates are not independent but are given exactly by the model evaluated at the adjusted  $x_{\text{true},i}$ . In this way, ODR enforces that each idealized point lies *exactly* on the fitted curve while minimizing the orthogonal distances between the measured and idealized points.

The goal of the algorithm is to find the model parameters ( $A, B$ ) that require the smallest overall adjustment to move the measured points ( $x_i, y_i$ ) to their corresponding idealized points ( $x_{\text{true},i}, y_{\text{true},i}$ ). This adjustment is quantified by the  $\chi^2$  function, which sums the weighted squared displacements in both the horizontal and vertical directions [7]:

$$\chi^2 = \sum_{i=1}^N \left[ \left( \frac{\epsilon_i}{\delta x_i} \right)^2 + \left( \frac{\delta_i}{\delta y_i} \right)^2 \right], \quad (16)$$

where  $\epsilon_i = x_i - x_{\text{true},i}$  and  $\delta_i = y_i - y_{\text{true},i}$  denote the horizontal and vertical residuals, respectively. By minimizing Eq. (16), ODR incorporates the full uncertainty information in both axes and thus provides the most statistically reliable estimates of the model parameters and their uncertainties.

## D. Period vs. Length Analysis

### 1. Length Measurement and Error

The pendulum length  $L$  was measured from the pivot point to the center of mass of the bob. This was done in two parts: first, measuring the length  $L_1$  from the pivot to the top of the bob, and second, measuring the length  $L_2$  from the top of the bob to its geometric center (assumed to be its center of mass). The total length is  $L = L_1 + L_2$ .

Both measurements were made with a meter stick with an uncertainty of 0.01 m. Adding these two independent measurements in quadrature gives the total error in length  $\delta L$ :

$$\delta L = \sqrt{(\delta L_1)^2 + (\delta L_2)^2} = \sqrt{2(0.01 \text{ m})^2} \approx 0.014 \text{ m} \quad (17)$$

[4]

## 2. Parameter Fitting

For the analysis of period versus length (performed in the small-angle regime) the data was fit directly to the non-linear power-law model  $T = C \cdot L^P$ . Both the constant  $C$  and the exponent  $P$  were treated as free parameters in the fit.

As established, the independent variable  $L$  has a non-zero uncertainty  $\delta L$ , and the dependent variable  $T$  has uncertainty  $\delta T$ . Standard fitting routines that assume zero uncertainty in the independent variable are insufficient. Therefore, the `scipy.odr` package was used[7] to perform a non-linear Orthogonal Distance Regression. This method correctly accounts for errors in both  $T$  and  $L$  by minimizing the weighted orthogonal distances from the data points to the fitted curve. This ensured that the uncertainties in both variables were properly incorporated into the fit for the two parameters,  $C$  and  $P$ .

## III. RESULTS

We first discuss the results of the  $f(\theta)$  experiments. Recall the general equation for the period, as derived from our model in Eq. (6):

$$T = \sqrt{\frac{l}{g}}(A + B\theta^2) \quad (18)$$

Dividing both sides by  $\sqrt{l/g}$  gives an equation for the "normalized period," which is equivalent to our function of interest:

$$f(\theta) = A + B\theta^2 = \frac{T}{\sqrt{l/g}}[1] \quad (19)$$

We hypothesized that this function is universal for all simple pendulums, regardless of length or mass. The following analysis tests this hypothesis.

Pictured in the page below are four plots: three for pendulums of the same mass but different lengths, and one plot comparing two pendulums of different mass but similar length.

### A. Weighted Mean Analysis

To rigorously test whether our experimental results agree with the theoretical predictions (found in the The-

ory section above), we combine the five independent measurements of parameters  $A$  and  $B$  using inverse-variance weighting. This approach gives more weight to precise measurements and less weight to those with larger uncertainties.

The weighted mean of a parameter  $X$  is calculated as:

$$\bar{X}_{\text{weighted}} = \frac{\sum_{i=1}^N w_i X_i}{\sum_{i=1}^N w_i} \quad (20)$$

[9] where the weights are given by the inverse variance:  $w_i = 1/\sigma_{X_i}^2$ . The uncertainty in the weighted mean is:

$$\sigma_{\bar{X}} = \sqrt{\frac{1}{\sum_{i=1}^N w_i}} \quad (21)$$

### 1. Analysis of Parameter A

For the five measurements of  $A$ , we calculate the weights as shown in Table II.

TABLE II. Weights for Parameter  $A$  Measurements

| Length (cm) | $A_i$ | $\sigma_{A_i}$ | $w_i = 1/\sigma_{A_i}^2$ |
|-------------|-------|----------------|--------------------------|
| 16.8        | 6.289 | 0.012          | 6944                     |
| 26.4        | 6.282 | 0.005          | 40000                    |
| 34.7        | 6.282 | 0.004          | 62500                    |
| 51.0 (37g)  | 6.290 | 0.005          | 40000                    |
| 50.8 (6g)   | 6.285 | 0.011          | 8265                     |
| $\sum w_i$  |       |                | 157709                   |

Applying Equation (20), we obtain:

$$\bar{A}_{\text{weighted}} = 6.2850 \pm 0.0025 \quad (22)$$

The theoretical prediction is  $A_{\text{theory}} = 2\pi = 6.2832$ . The deviation from theory is:

$$z_A = \frac{\bar{A}_{\text{weighted}} - A_{\text{theory}}}{\sigma_{\bar{A}}} = \frac{6.2850 - 6.2832}{0.0025} = 0.72 \sigma \quad (23)$$

Since  $|z_A| < 2$ , our experimental value of  $A$  agrees with the theoretical prediction within one standard deviation, strongly confirming that the zeroth-order term in the normalized period is indeed  $2\pi$ .

### 2. Analysis of Parameter B

Similarly, for parameter  $B$ , the weights and weighted mean are calculated as shown in Table III.

The weighted mean is:

$$\bar{B}_{\text{weighted}} = 0.3973 \pm 0.0102 \text{ rad}^{-2} \quad (24)$$

The theoretical prediction is  $B_{\text{theory}} = 2\pi/16 = 0.3927 \text{ rad}^{-2}$ . The deviation is:

$$z_B = \frac{\bar{B}_{\text{weighted}} - B_{\text{theory}}}{\sigma_{\bar{B}}} = \frac{0.3973 - 0.3927}{0.0102} = 0.45 \sigma \quad (25)$$

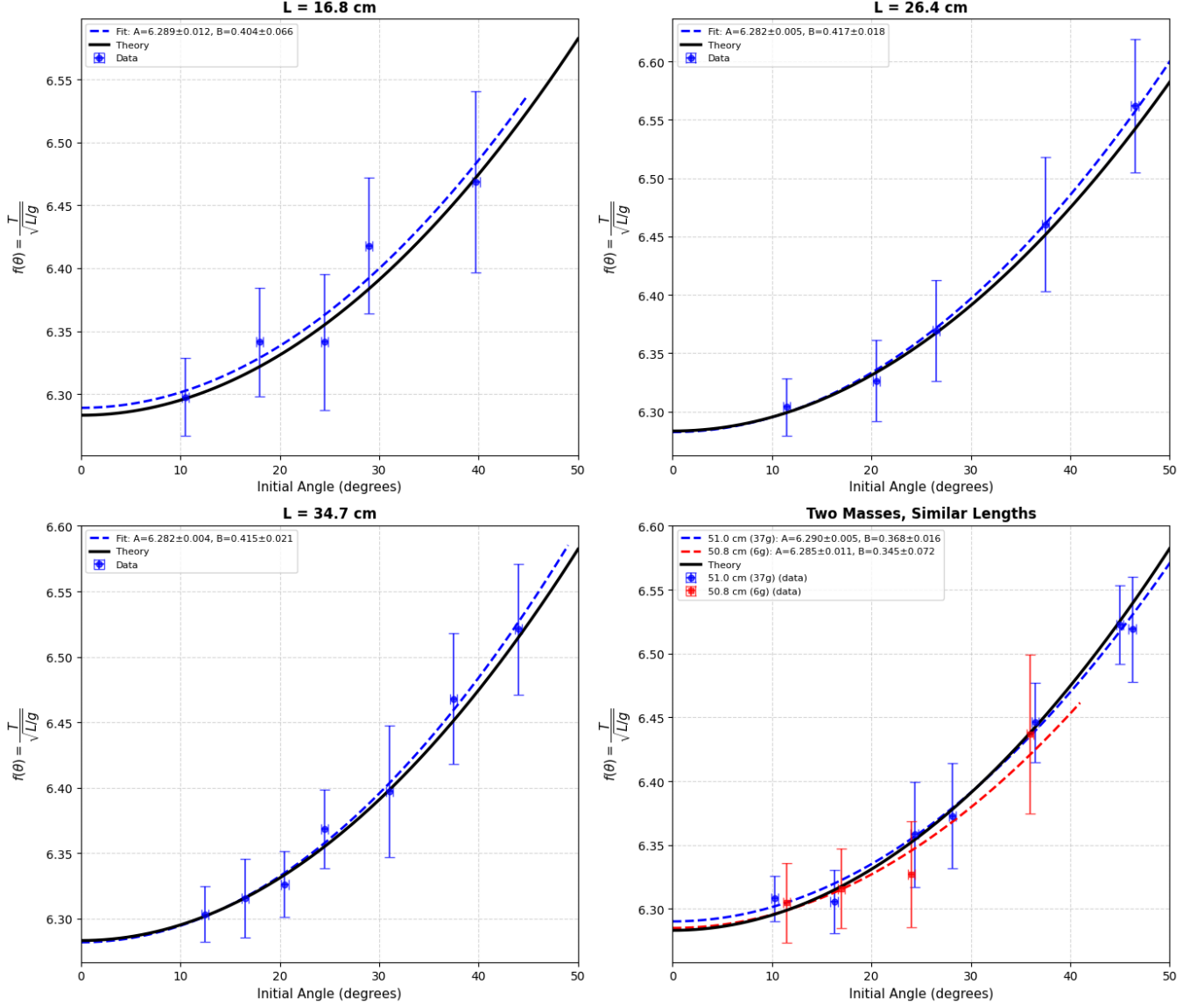


FIG. 3. Four plots of  $f(\theta)$ . Note: for most data points, one measurement of period was taken.

TABLE III. Weights for Parameter  $B$  Measurements

| Length (cm) | $B_i$ ( $\text{rad}^{-2}$ ) | $\sigma_{B_i}$ ( $\text{rad}^{-2}$ ) | $w_i = 1/\sigma_{B_i}^2$ |
|-------------|-----------------------------|--------------------------------------|--------------------------|
| 16.8        | 0.404                       | 0.066                                | 230                      |
| 26.4        | 0.417                       | 0.018                                | 3086                     |
| 34.7        | 0.415                       | 0.021                                | 2268                     |
| 51.0 (37g)  | 0.368                       | 0.016                                | 3906                     |
| 50.8 (6g)   | 0.345                       | 0.072                                | 193                      |
| $\sum w_i$  |                             |                                      | 9683                     |

This result also shows excellent agreement with theory within one standard deviation.

## B. Period vs. Length

## C. Test of Mass Independence

A critical assumption of pendulum theory is that the period (and hence the normalized period function) is independent of the bob mass. To test this directly, we compare the results from two pendulums of nearly identical length but vastly different masses: 51.0 cm with 37 g versus 50.8 cm with 6 g. Since the maximum velocity scales as  $v_{\text{max}} \propto \sqrt{gL}$ , the longest pendulum experiences the highest velocities and thus the largest air resistance forces. If mass-dependent effects from air drag were to manifest anywhere in our experimental range, they would be most pronounced here. By demonstrating mass independence even in this extreme case—with a mass ratio

of approximately 6:1—we can confidently conclude that mass effects are negligible across all our pendulum configurations.

### 1. Mass Independence of Parameter $A$

The difference in  $A$  values between the two pendulums of similar length is:

$$\Delta A = |A_{37g} - A_{6g}| = |6.290 - 6.285| = 0.005 \quad (26)$$

The combined uncertainty is:

$$\sigma_{\Delta A} = \sqrt{\sigma_{A,37g}^2 + \sigma_{A,6g}^2} = \sqrt{0.005^2 + 0.011^2} = 0.012 \quad (27)$$

Since the combined experimental uncertainty ( $\sigma_{\Delta A} = 0.012$ ) is approximately twice as large as the observed difference ( $\Delta A = 0.005$ ), the two measurements are statistically indistinguishable. This confirms that mass does not affect parameter  $A$  within our experimental precision.

### 2. Mass Independence of Parameter $B$

Similarly, for parameter  $B$ , the difference between the two pendulums is:

$$\Delta B = |B_{37g} - B_{6g}| = |0.368 - 0.345| = 0.023 \text{ rad}^{-2} \quad (28)$$

The combined uncertainty is:

$$\sigma_{\Delta B} = \sqrt{\sigma_{B,37g}^2 + \sigma_{B,6g}^2} = \sqrt{0.016^2 + 0.072^2} = 0.074 \text{ rad}^{-2} \quad (29)$$

Since the combined experimental uncertainty ( $\sigma_{\Delta B} = 0.074 \text{ rad}^{-2}$ ) is more than three times larger than the observed difference ( $\Delta B = 0.023 \text{ rad}^{-2}$ ), the two measurements are statistically indistinguishable. This confirms that mass does not affect parameter  $B$  within our experimental precision.

## D. Discussion of Systematic Effects

While our weighted mean value of  $B$  agrees statistically with theory (within  $0.45\sigma$ ), the spread in the five individual measurements of  $B$  is noticeably larger than the spread in the measurements of  $A$ . Examining Tables II and III, we see that  $B$  ranges from 0.345 to  $0.417 \text{ rad}^{-2}$  (a span of  $0.072 \text{ rad}^{-2}$ , or about 18% relative variation), whereas  $A$  ranges only from 6.282 to 6.290 (a span of 0.008, or about 0.13% relative variation). This difference in consistency can be explained by the physical nature of the two parameters.

Parameter  $A$  represents the zeroth-order (dominant) term in the normalized period function, contributing the entire value at small angles and remaining the dominant contribution even at moderate angles. In contrast,

parameter  $B$  represents the first-order correction term, which, for example, contributes only about 6% to the total period at  $30^\circ$  (approximately  $0.39 \times (0.52)^2 \approx 0.11$  compared to  $A \approx 6.28$ ).

Because  $B$  is a small correction, it is inherently more sensitive to experimental noise, with random fluctuations in period measurements having a proportionally larger effect on the fitted value of  $B$  than on  $A$ . Additionally, different pendulum configurations were measured over different angular ranges, and since the quadratic approximation becomes less accurate at larger angles, the effective value of  $B$  extracted from the fit may depend on the specific angles sampled. Furthermore, with fewer data points at larger angles where the  $B\theta^2$  term becomes more significant, the statistical power to constrain  $B$  is reduced compared to the dominant parameter  $A$ .

Despite this variability, all five measurements of  $B$  remain within approximately  $2\sigma$  of each other and of the theoretical value, confirming the validity of the first-order correction term.

## E. Analysis of Period vs. Length ( $T_0$ vs. $L$ )

The second goal of this experiment was to confirm the power-law relationship between the pendulum's length  $L$  and its period. To remain consistent across the five pendulums tested, it would not make sense to average all period measurements. Instead, we use the period measurement in the small-angle regime (i.e., when the  $\theta^2$  term in our model can be neglected). To acquire the data for this fit, we used the results from the  $f(\theta)$  analysis in the previous section.

The small-angle period  $T_0$  (the period at  $\theta = 0$ ) is simply the normalized small-angle period  $f(0) = A$ , multiplied by the pendulum's specific normalization factor. Thus, for each pendulum,  $T_0$  was calculated as:

$$T_0 = A \cdot \sqrt{\frac{L}{g}} \quad (30)$$

The uncertainty for this  $T_0$  was then propagated[4] from the uncertainty in the fitted intercept,  $\delta A$ , and the uncertainty in the measured length,  $\delta L$ . This process yielded five independent data points ( $L \pm \delta L, T_0 \pm \delta T_0$ ).

This dataset was then used to test the theoretical power-law  $T_0 = C \cdot L^P$ . The plot shown in Figure 4 demonstrates that the data points, with their error bars, lie essentially on the fitted line. The experimental model obtained is

$$T_0 = (2.009 \pm 0.002) \cdot L^{(0.501 \pm 0.001)}. \quad (31)$$

### 1. Analysis of the Exponent $P$

To assess agreement with theory, we compare the fitted exponent  $P = 0.501 \pm 0.001$  with the theoretical value

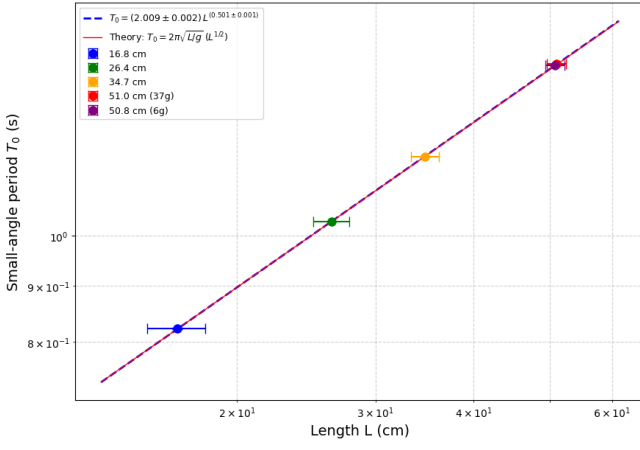


FIG. 4. Small-angle period  $T_0$  vs pendulum length  $L$ , plotted on a log-log scale to illustrate the power-law relationship. The solid red line shows the theoretical  $L^{0.5}$  relationship, while the dashed blue line (which overlaps) shows the ODR fit  $L^{0.501}$ . The lines are nearly indistinguishable.

$P_{\text{theory}} = 0.5$ . The deviation in units of standard deviations is

$$z_P = \frac{P_{\text{exp}} - P_{\text{theory}}}{\delta P_{\text{exp}}} = \frac{0.501 - 0.500}{0.001} = 1.0 \sigma. \quad (32)$$

Since  $|z_P| < 2$ , the exponent is in excellent agreement with theory, confirming that  $T_0 \propto L^{0.5}$  within experimental uncertainty.

## 2. Analysis of the Constant $C$

Similarly, the fitted constant  $C = 2.009 \pm 0.002 \text{ s/m}^{0.5}$  can be compared to the theoretical value

$$C_{\text{theory}} = \frac{2\pi}{\sqrt{g}} \approx 2.0064 \text{ s/m}^{0.5}[1]. \quad (33)$$

The deviation in standard deviations is

$$z_C = \frac{C_{\text{exp}} - C_{\text{theory}}}{\delta C_{\text{exp}}} = \frac{2.009 - 2.0064}{0.002} = 1.3 \sigma. \quad (34)$$

This confirms that the measured constant is consistent with theory.

Overall, both the exponent  $P$  and the constant  $C$  demonstrate excellent agreement with the theoretical model  $T_0 = 2\pi\sqrt{L/g}$ , with deviations well within one to two standard deviations. This provides strong quantitative support for the expected power-law dependence of  $T_0$  on  $L$ .

As a final test of the model, all normalized period data from the five pendulums were plotted together in Figure 5 below, providing a clear visual summary of the results.

The combined data collapse onto a single curve, despite the pendulums varying in length by a factor of three (16.8 cm to 51.0 cm) and in mass by a factor of six (6 g

to 37 g). This observation quantitatively confirms that the normalized period  $f(\theta)$  is independent of both mass and length, providing strong support for the theoretical prediction.

## F. Summary of Results

Table IV summarizes the key statistical results.

TABLE IV. Summary of Statistical Analysis

| Parameter                     | Weighted Mean                                                     | Theory                     | Deviation     |
|-------------------------------|-------------------------------------------------------------------|----------------------------|---------------|
| $A$                           | $6.2850 \pm 0.0025$                                               | 6.2832                     | $0.72 \sigma$ |
| $B$                           | $0.3973 \pm 0.0102 \text{ rad}^{-2}$                              | $0.3927 \text{ rad}^{-2}$  | $0.45 \sigma$ |
| Mass Independence Test        |                                                                   |                            |               |
| $\Delta A$                    | 0.005 (uncertainty: 0.012)                                        |                            |               |
| $\Delta B$                    | $0.023 \text{ rad}^{-2}$ (uncertainty: $0.074 \text{ rad}^{-2}$ ) |                            |               |
| Length Dependence (Power Law) |                                                                   |                            |               |
| $P$ (exponent)                | $0.501 \pm 0.001$                                                 | 0.5000                     | $1.0 \sigma$  |
| $C$ (constant)                | $2.009 \pm 0.002 \text{ s/m}^{0.5}$                               | $2.0064 \text{ s/m}^{0.5}$ | $1.3 \sigma$  |

## G. Conclusions

Our experiments and data analysis strongly support the theoretical predictions for the nonlinear pendulum at moderate angles. The weighted mean of our five independent measurements yields  $A = 6.2850 \pm 0.0025$ , deviating from the theoretical value of  $2\pi$  by only  $0.72 \sigma$ . This excellent agreement, combined with the high consistency among all five measurements, strongly confirms that the zeroth-order term in the normalized period is indeed  $2\pi$ .

For the first-order coefficient, we obtained  $B = 0.3973 \pm 0.0102 \text{ rad}^{-2}$ , which agrees with the theoretical value of  $2\pi/16 = 0.3927 \text{ rad}^{-2}$  to within  $0.45 \sigma$ . While the individual measurements of  $B$  show greater variability than those of  $A$ , this is expected since  $B$  represents a small correction term making it inherently more sensitive to experimental noise and fitting uncertainties.

We directly verified mass independence by comparing two pendulums of nearly identical length (51.0 cm and 50.8 cm) but vastly different masses (37 g and 6 g). The combined experimental uncertainties were approximately twice as large as the observed difference for parameter  $A$  and more than three times larger for parameter  $B$ , confirming that mass does not affect the normalized period function within our experimental precision.

The power-law relationship between period and length was confirmed with exceptional precision. Our experimental exponent  $P = 0.501 \pm 0.001$  deviates from the theoretical value of  $0.5$  by only  $1.0 \sigma$ , providing definitive confirmation of the  $T_0 \propto \sqrt{L}$  relationship. Similarly, the fitted constant  $C = 2.009 \pm 0.002 \text{ s/m}^{0.5}$  agrees with the theoretical prediction  $C = 2\pi/\sqrt{g} = 2.0064 \text{ s/m}^{0.5}$  to within  $1.3 \sigma$ . The precision of these measurements—with relative uncertainties below  $0.2\%$  for the exponent and

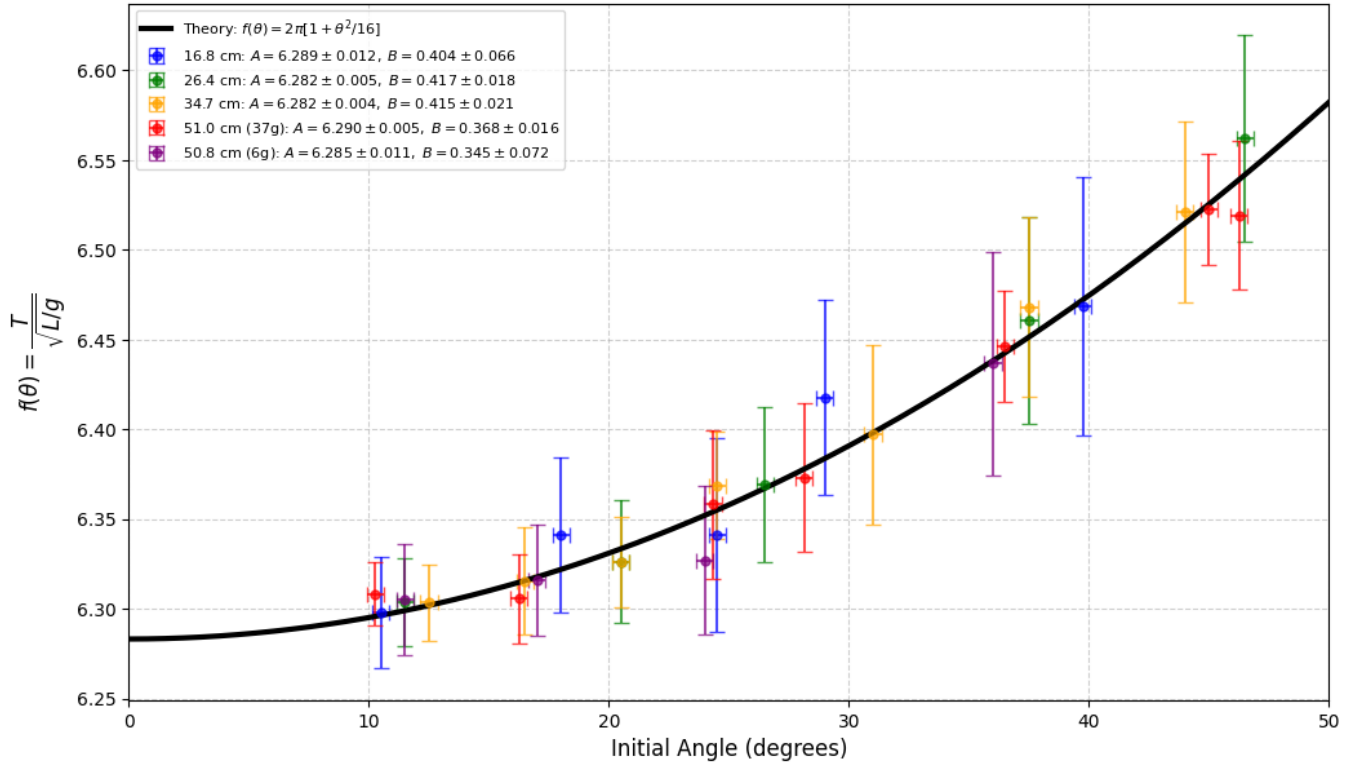


FIG. 5. Normalized period  $f(\theta) = T/\sqrt{L/g}$  versus initial angle  $\theta$ . Data from all five pendulums are plotted together. The solid black line shows the theoretical prediction  $f_{\text{theory}}(\theta) = 2\pi\left(1 + \frac{\theta^2}{16}\right)$ , demonstrating excellent agreement with the combined experimental data.

0.1% for the constant—demonstrates both the validity of the theoretical model and the integrity of our experimental method. Together, these results provide comprehensive experimental confirmation of the theoretical model for the nonlinear pendulum at moderate angles, validating both the angular dependence and the scaling relationship.

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